Traditional Machine Learning Documentation

KIO ROUTER AI • IMPLEMENTATION, EVALUATION, & SYSTEM INTEGRATION

Item	Recorded Status
Repository & Branch	`omardiab9951/KIO-Router-AI` (branch: `feature/traditional-ml`)
Recorded Commits	`3fd2f69` (ML pipeline), `2714aa4` (test evaluation results)
Latest Changes Status	**[CONFIRM: committed or not, and to which branch]**
Final Test Accuracy	**90% (18/20 correct)**; initial baseline was 60% (12/20)
Automated Unit Tests	5 passed (in `tests/test_ml_router.py`)

1. Project Overview

The Traditional ML component converts raw customer text into numerical vector representations and classifies each query into either **Support (Flag 0)** or **Sales (Flag 1)** using statistical word-pattern learning. It is compared directly against Cloudflare Clef (Decision AI), which operates on zero-shot instruction parsing without dataset training.

2. ML Pipeline Architecture

```
Customer message → TF-IDF (words + word pairs) → Logistic Regression → Support (0) / Sales (1)
```

Stage	Purpose	Notes & Design Rationale
Input Text	Receive message string	e.g., *"I can't access my account online."*
TF-IDF Vectorizer	Convert text to numerical matrix	Extracts unigrams and bigrams using sublinear term frequency scaling.
Logistic Regression	Predict classification flag	Linear classifier; regularization parameter C tuned via 5-fold CV.

* Design refinement: An earlier version compressed features into 2D using Truncated SVD, causing severe information loss (60% accuracy). Removing SVD from predictions raised accuracy to 90%. SVD is now used solely for 2D visualization plots.

3. Model Hyperparameters & Settings

Setting	Value	Explanation & Impact
`lowercase`	`True`	Normalizes case ("Internet" vs "internet").
`ngram_range`	`(1, 2)`	Captures both single words and key pairs like "not interested".
`sublinear_tf`	`True`	Sublinear scaling prevents high word counts from dominating weights.
`C`	**0.1**	Regularization strength selected via grouped CV on training split.
`max_iter` / `random_state`	2000 / 42	Ensures optimizer convergence and repeatable experiments.

4. System Integration & Architecture

- **Class Wrapper (`kio/ml_router.py`):** Implements `MLRouter`, loading the saved scikit-learn pipeline (`models/ml_model.joblib`) once and returning a standardized `RouterOutput`.
- **Validation & Fallback:** Empty input messages or pipeline missing states pass through unified validation to yield an `invalid` output rather than making false guesses.
- **Dispatcher Integration:** Managed via `kio/router.py` alongside Clef and Mock modes, serving responses directly to the Streamlit frontend.

5. Evaluation Results (20 Frozen Test Messages)

Metric	Final Result	Initial Baseline	Interpretation
Correct / Incorrect	**18 / 2**	12 / 8	18 predictions match true labels
Accuracy	**90%**	60%	Each test message represents 5% accuracy
Confusion Matrix	[[10, 0], [2, 8]]	[[8, 2], [6, 4]]	10 Support correct, 8 Sales correct, 2 Sales missed
Sales Precision / Recall / F1	100% / 80% / 0.889	66.7% / 40% / 0.500	Zero false sales positives; 2 recall misses
Average Latency	**~8.8 ms**	N/A	Extremely fast local inference

6. Error Analysis & Model Disagreements

The ML model achieved 90% accuracy, making errors exclusively on two Sales messages featuring heavy negation:

- **ID 59:** *"I am not interested in TV, just give me the internet prices."* (Predicted: Support | True: Sales)
- **ID 60:** *"No TV channels, just pure fiber broadband, how much?"* (Predicted: Support | True: Sales)

Root Cause: A bag-of-words / n-gram representation detects negative terms ("not", "no") but fails to resolve syntactic scope, mistakenly associating negation tokens with customer complaints or support issues.

7. Comparative Analysis: Traditional ML vs. Clef AI

Dimension	Traditional ML	Cloudflare Clef (Decision AI)
Accuracy (20 Test Messages)	90% (18/20)	**100% (20/20)**
Inference Speed (Latency)	**~8.8 ms** (170x faster)	~1,467 ms
Dependency & Cost	Offline / Zero cost	Requires Internet & API Key
Negation Handling	Fails on complex negation	Handles complex negation accurately

8. Limitations & Recommendations

- **Data Volume Caveat:** With 20 test messages, each error alters accuracy by 5 percentage points. Cross-validated accuracy (82.5%) and grid-tuning history suggest a realistic operational accuracy between **85% and 90%**.
- **Hybrid Architecture Recommendation:** Use the local ML router as a high-speed primary filter. Route requests to Clef when ML confidence is low (near 0.50) or when negation keywords are detected.

9. Command Reference

```
python -m pytest
python ml_model/evaluate_ml_model.py --dry-run
python comparison/compare_results.py
python -m streamlit run frontend/app.py
```